

ISHIGAKIJIMA, Japan

WMO: 479180

Lat: 24.3367N

Lon: 124.1644E

Elev: 15

StdP: 101.15

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.9	13.9	4.5	5.2	14.3	5.8	5.7	15.1	10.8	19.2	10.0	19.3	5.4	20	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.8	32.6	27.1	32.2	27.0	31.7	26.8	28.2	31.0	27.9	30.7	27.6	30.5	5.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.4	23.3	30.1	27.1	22.9	30.0	26.7	22.4	29.8	90.9	31.2	89.3	30.8	87.9	30.5	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
12.3	10.1	9.0	DB	11.6	33.7	1.2	0.6	10.7	34.1	10.0	34.4	9.3	34.8	8.4	35.2
			WB	7.8	28.3	1.4	0.6	6.8	28.7	6.0	29.0	5.3	29.4	4.3	29.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	19.0	19.5	20.9	23.5	26.1	28.6	29.8	29.5	28.4	26.3	23.8	20.6
	DBStd	4.32	2.59	2.88	2.69	2.23	1.80	1.57	1.04	1.05	1.36	1.59	2.04	2.55
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	66	24	21	9	1	0	0	0	0	0	0	1	10
	CDD10.0	5364	278	267	339	406	500	557	613	604	552	505	414	329
	CDD18.3	2388	44	54	91	157	242	307	355	346	302	247	164	81
	CDH23.3	22412	39	77	222	768	2054	3622	4661	4449	3492	2077	802	148
CDH26.7	7757	0	1	3	52	404	1381	2190	1985	1253	418	68	2	
Wind	WSAvg	5.0	5.1	5.1	4.8	4.6	4.3	5.0	5.4	5.1	5.0	5.2	5.2	5.3
Precipitation	PrecAvg	2073	128	117	130	145	207	201	162	241	243	193	163	139
	PrecMax	3374	358	578	377	480	638	688	630	617	826	934	448	462
	PrecMin	1114	32	12	24	6	6	13	8	26	30	17	17	20
	PrecStd	455	70	84	79	115	145	135	137	139	161	176	94	98
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.6	26.2	27.2	28.9	31.0	32.6	33.3	33.3	32.7	31.3	29.3	26.8
		MCWB	21.3	22.7	23.7	25.2	26.8	27.8	27.3	27.0	26.8	26.3	24.9	22.6
	2%	DB	24.4	25.1	26.1	28.0	30.1	31.8	32.6	32.6	31.9	30.3	28.3	25.7
		MCWB	20.5	21.6	22.7	24.5	26.1	27.4	27.2	26.9	26.5	25.5	24.1	21.7
	5%	DB	23.6	24.3	25.4	27.2	29.4	31.2	32.2	32.1	31.3	29.5	27.5	24.9
		MCWB	19.8	21.0	22.1	24.1	25.7	27.1	27.0	26.8	26.2	24.9	23.4	21.2
	10%	DB	22.7	23.5	24.6	26.5	28.7	30.7	31.7	31.5	30.7	28.8	26.7	24.1
		MCWB	19.2	20.3	21.5	23.6	25.3	26.9	26.9	26.7	25.9	24.4	22.8	20.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	23.5	24.3	26.1	27.6	28.6	28.6	28.6	28.0	27.2	25.7	24.0
		MCDB	24.2	25.5	26.3	28.0	29.9	31.3	31.8	31.2	31.3	29.8	28.2	25.6
	2%	WB	21.5	22.6	23.6	25.4	26.9	28.1	28.1	28.1	27.5	26.6	25.1	22.9
		MCDB	23.5	24.5	25.5	27.2	29.1	30.6	31.2	31.0	30.6	29.2	27.4	24.9
	5%	WB	20.7	21.8	22.9	24.8	26.4	27.7	27.7	27.7	27.0	26.0	24.5	22.0
		MCDB	22.9	23.8	24.9	26.6	28.5	30.2	30.8	30.7	30.1	28.6	26.7	24.2
	10%	WB	19.8	20.9	22.2	24.2	25.9	27.4	27.5	27.2	26.6	25.4	23.8	21.0
		MCDB	22.3	23.1	24.3	26.0	28.1	30.0	30.6	30.4	29.7	28.1	26.1	23.4
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.3	4.5	4.2	4.1	3.6	3.8	4.0	4.3	4.3	4.1	4.1
		MCDBR	4.8	4.7	4.7	4.2	4.2	3.7	4.1	4.5	4.7	4.7	4.5	4.6
		MCWBR	3.1	3.2	3.3	2.6	2.3	1.7	1.7	1.8	2.0	2.2	2.5	3.0
	5% WB	MCDBR	4.3	4.4	4.4	3.8	3.9	3.6	3.8	4.0	4.2	4.3	4.0	4.1
		MCWBR	3.4	3.6	3.9	2.7	2.3	1.8	1.8	1.9	2.1	2.2	2.6	3.5
Clear-Sky Solar Irradiance	taub	0.488	0.507	0.595	0.600	0.522	0.486	0.481	0.499	0.495	0.499	0.491	0.495	
	taud	2.074	2.052	1.850	1.891	2.160	2.298	2.309	2.245	2.228	2.161	2.151	2.062	
	Ebn at Noon	757	776	728	735	791	814	818	805	801	774	751	730	
	Edh at Noon	148	161	206	202	154	133	131	140	140	143	136	145	
All-Sky Solar Radiation	RadAvg	2.37	3.03	3.67	4.34	5.13	5.78	6.43	5.82	5.16	4.10	2.94	2.32	
	RadStd	0.37	0.54	0.47	0.56	0.60	0.46	0.42	0.42	0.56	0.44	0.33	0.32	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.26	N/A	N/A	+0.33	+0.58	+0.69	N/A	N/A	+98	+90
Regional (3 neighbors)		+0.23	N/A	N/A	+0.31	+0.43	+0.46	N/A	N/A	+75	+69

Nomenclature: See separate page